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### VESSEL RINSING APPARATUS

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#### Abstract

A vessel rinsing apparatus including a drain bowl, a fluid discharge control including trigger and a sprayface having an outlet. A valve is operably coupled to the fluid discharge control and configured to control water flow through the outlet, and a lock to prevent movement of the trigger to a lowered position.

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#### Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] The present application claims priority to U.S. Provisional Patent Application Ser. No. 63/557,460, filed Feb. 23, 2024, the disclosure of which is expressly incorporated herein by reference.

## BACKGROUND AND SUMMARY OF THE DISCLOSURE

[0002] The present invention relates generally to a fluid dispensing system and, more particularly, to a vessel rinsing apparatus for use with containers, such as fluid vessels or receptacles, including drinkware (e.g., cups, bottles and/or glasses).

[0003] Glass rinsers are known in the art to provide for the cleaning of various containers.

However, there is a need for a vessel rinser that may be mounted to a sink deck for draining into a sink basin and including a lockout mechanism to prevent inadvertent actuation thereof.

[0004] According to an illustrative embodiment of the present disclosure, a vessel rinsing apparatus includes a hollow, cylindrical mounting shank extending along a longitudinal axis, the mounting shank defining a first fluid passageway. A valve includes a valve seat supported by the mounting shank, a valve stem movable along the longitudinal axis, and a valve seal supported by the valve stem. The valve stem is movable between a closed position and an open position. The valve seal is bias against the valve seat in the closed position. A fluid discharge control includes a central body extending along the longitudinal axis and at least partially disposed within the mounting shank, wherein the central body includes a second fluid passageway. A sprayface is supported by a first end of the central body and includes at least one outlet in fluid communication with the second fluid passageway. A trigger is operably coupled to the central body, wherein the trigger is movable for axial movement along the longitudinal axis between a raised position and a lowered position. The trigger is movable for rotational movement about the longitudinal axis between a first rotation position and a second rotational position. The trigger is biased towards the raised position. Moving the trigger to the lowered position is configured to move the valve seal away from the valve seat causing the valve to be in the open position. A lock is configured to prevent movement of the trigger to the lowered position when the trigger is in the first rotational position. A drain bowl is operably coupled to the mounting shank, the drain bowl being defined by a base and an upwardly extending wall.

[0005] According to another illustrative embodiment of the present disclosure, a drinkware rinsing apparatus includes a drain bowl defined by a base and an upwardly extending wall. A valve includes a valve seat supported by the mounting shank, a valve stem movable along the longitudinal axis, and a valve seal supported by the valve stem. The valve stem is movable between a closed position and an opened position. The valve seal is biased against the valve seat in the closed position. A fluid discharge control includes a central body extending along the longitudinal axis and at least partially disposed within the mounting shank, wherein the central body includes an outlet fluid passageway. A sprayface is supported by a first end of the central body and includes at least one outlet in fluid communication with the fluid passageway. The sprayface is located substantially within the drain bowl. A trigger is operably coupled to the central body, wherein the trigger is movable for axial movement along the longitudinal axis between a raised position and a lowered position. The trigger is biased towards the raised position. The trigger is movable for rotational movement about the longitudinal axis between a first rotational position and a second rotation position. The trigger is biased toward the first rotational position. Moving the trigger to the lowered position is configured to move the valve seal away from the valve seat causing the valve to be in the open position. A lock is configured to prevent movement of the trigger to the lowered position when the trigger is in the first rotational position.

[0006] According to a further illustrative embodiment of the present disclosure, a vessel rinsing apparatus includes a hollow, cylindrical mounting shank extending along the longitudinal axis, the mounting shank defining the first fluid passageway. A valve includes a fixed valve member supported within the mounting shank and defining a valve seat, a valve stem movable along the

longitudinal axis, and a valve seal supported by the valve stem. The valve stem is movable between a closed position and an open position. The valve seal is bias against the valve seat in the closed position. A fluid discharge control includes a central body extending along the longitudinal axis and at least partially disposed within the mounting shank, wherein the central body includes a second fluid passageway. A sprayface is supported by a first end of the central body and includes at least one outlet in fluid communication with the second fluid passageway. A trigger is operably coupled to the central body. The trigger is movable for axial movement along the longitudinal axis between a raised position and lowered position. The trigger is movable for rotational movement about the longitudinal axis between a first rotational position and a second rotational position. The trigger is bias towards the raised position. Moving the trigger to the lowered position is configured to move the valve seal away from the valve seat causing the valve to be in the open position. The lock is configured to prevent movement of the trigger to the lowered position when the trigger is in the first rotational position. A rotatable drive is operably coupled to the central body. An engagement surface is supported by the rotatable drive and is configured to move the valve stem toward the open position when the trigger is in the lowered position. A drain bowl is operably coupled to the mounting shank, the drain bowl being defined by a base and an upwardly extending wall. A drain channel extends through the upwardly extending wall of the drain bowl. In an installed condition of the rinsing apparatus, the drain bowl is positioned above the mounting deck, at least a portion of the drain channel is positioned above the sink basin, and the drain channel is configured to receive fluid from the drain bowl and discharge the fluid above and into the sink basin.

[0007] Additional features and advantages of the present invention will become apparent to those skilled in the art upon consideration of the following detailed description of the illustrative embodiment exemplifying the best mode of carrying out the invention as presently perceived.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0008] A detailed description of the drawings particularly refers to the accompanying figures, in which:

[0009] FIG. 1 is a perspective view of an illustrative vessel rinsing apparatus mounted to a sink deck and fluidly coupled to a faucet;

[0010] FIG. 2 is a perspective view of the illustrative vessel rinsing apparatus of FIG. 1;

[0011] FIG. 3 is a perspective view of the vessel rinsing apparatus of FIG. 2, showing the quick connect coupler uncoupled from the mounting shank;

[0012] FIG. 4 is an exploded perspective view of the vessel rinsing apparatus of FIG. 1;

[0013] FIG. 5 is a cross-sectional view taken along line 5-5 of FIG. 2, showing the quick connect coupler of the vessel rinsing apparatus;

[0014] FIG. 6A is a cross-sectional view taken along line 6-6 of FIG. 3, showing a glass positioned above the trigger of the vessel rinsing apparatus;

[0015] FIG. 6B is a partial cross-sectional view of the vessel rinsing apparatus of FIG. 6A, showing the valve in a closed position;

[0016] FIG. 6C is a top plan view of the vessel rinsing apparatus of FIG. 6A;

[0017] FIG. 7A is a cross-sectional view taken along line 6-6 of FIG. 3, showing a glass depressing the trigger of the vessel rinsing apparatus;

[0018] FIG. 7B is a partial cross-sectional view of the vessel rinsing apparatus of FIG. 7A, showing the valve in an open position;

[0019] FIG. 7C is a top plan view of the vessel rinsing apparatus of FIG. 7A;

[0020] FIG. 8 is a partial perspective view, in cross-section, showing the trigger of the vessel rinsing apparatus of FIG. 1;

[0021] FIG. **9** is a perspective view of the trigger of FIG. **8**;  
[0022] FIG. **10** is a cross-sectional view taken along line **10-10** of FIG. **9**;  
[0023] FIG. **11A** is a cross-sectional view showing a first position of the trigger of the vessel rinsing apparatus of FIG. **2**;  
[0024] FIG. **11B** is a cross-sectional view showing a downward axial force applied to the trigger of FIG. **11A**;  
[0025] FIG. **11C** is a cross-sectional view showing a second position of the trigger of FIG. **11A** as a rotational force and a downward axial force is applied to the trigger;  
[0026] FIG. **12** is a perspective view of the mounting shank of the vessel rinsing apparatus of FIG. **2**;  
[0027] FIG. **13** is a perspective view of the rotatable drive of the trigger of the vessel rinsing apparatus of FIG. **2**;  
[0028] FIG. **14** is a perspective view of the fixed valve member of the vessel rinsing apparatus of FIG. **2**;  
[0029] FIG. **15** is a top plan view of an illustrative sink including a vessel rinsing apparatus of the present disclosure; and  
[0030] FIG. **16** is a partial perspective view of the illustrative sink of FIG. **15**.

#### DETAILED DESCRIPTION OF THE DRAWINGS

[0031] For the purposes of promoting and understanding the principles of the present disclosure, reference will now be made to the embodiments illustrated in the drawings, which are described herein. The embodiments disclosed herein are not intended to be exhaustive or to limit the invention to the precise form disclosed. Rather, the embodiments are chosen and described so that others skilled in the art may utilize their teachings. Therefore, no limitation of the scope of the claimed invention is thereby intended. The present invention includes any alterations and further modifications of the illustrated devices and described methods and further applications of principles in the invention which would normally occur to one skilled in the art to which the invention relates.

[0032] With reference initially to FIG. **1** of the drawings, an illustrative vessel rinsing apparatus or glass rinser **10** is shown supported by a conventional mounting deck, such as a sink deck **12** supporting a sink basin **14**. A faucet **16** may be supported by the sink deck **12** for discharging water from a water outlet **18** into the sink basin **14**. The faucet **16** may be of conventional design, including a delivery spout **20** defining the water outlet **18**, and a manual mixing valve **22** for controlling water flow from a hot water supply (e.g., a hot water stop) **24** and a cold water supply (e.g., a cold water stop) **26** to the water outlet **18**.

[0033] The mixing valve **22** may be of conventional design including a valve handle **28** to control the flow rate and the temperature of water delivered to the water outlet **18**. A hot water supply tube **30** may fluidly couple the hot water supply **24** to a first inlet of the mixing valve **22**, and a cold water supply tube **32** may fluidly couple the cold water supply **26** to a second inlet of the mixing valve **22**. A mixed water outlet passageway **34**, illustratively a tube received within the spout **20**, may fluidly couple an outlet of the mixing valve **22** to the outlet **18**.

[0034] An escutcheon **36** may be positioned intermediate the sink deck **12** and the delivery spout **20**. A plurality of mounting apertures **37** may extend within the sink deck **12**, wherein at least some of the apertures **37a**, **37b** may receive mounting shanks **38a**, **38b** used to couple the faucet **16** to the sink deck **12**. In the illustrative embodiment, the glass rinsing apparatus **10** is fixed to the mounting aperture **37c** that would conventionally support a kitchen side sprayer (not shown) adjacent the sink basin **14**. Such positioning facilitates draining of the vessel rinsing apparatus **10** into the sink basin **14**.

[0035] A fluid coupler **40**, illustratively a T-fitting, provides fluid communication between the cold water stop **26** and the faucet **16**, and between the cold water stop **26** and the vessel rinsing apparatus **10**. More particularly, the cold water supply tube **32** fluidly couples the T-fitting **40** and

the mixing valve **22** of the faucet **16**, while a fluid dispensing apparatus or vessel rinser supply tube **42** fluidly couples the T-fitting **40** and the vessel rinsing apparatus **10**. In other illustrative embodiments, a diverter valve (not shown) may be operably coupled between the vessel rinsing apparatus **10** and the faucet **16** to toggle water therebetween. Such a diverter valve arrangement is shown in U.S. Pat. No. 10,914,056 to Cipriani et al., the disclosure of which is expressly incorporated herein by reference.

[0036] With reference now to FIGS. 2-5, the illustrative vessel rinsing apparatus or glass rinser **10** includes a mounting shank **50** configured to extend through the sink deck **12**. The mounting shank **50** is hollow and extends vertically along a longitudinal axis **52** between a lower end **54** and an upper end **56**. More particularly, the illustrative mounting shank **50** includes a cylindrical side wall **58** having a plurality of lower external threads **60** to receive a mounting nut **62**. Cooperation between the mounting shank **50** and the mounting nut **62** couple the vessel rinsing apparatus **10** to an upper surface **64** of the sink deck **12** (FIG. 1).

[0037] With reference to FIGS. 1-6B, the mounting shank **50** defines a first fluid passageway **66** extending upwardly from the lower end **54**. As noted above, the vessel rinser supply tube **42** fluidly couples the cold water supply **26** to the fluid dispensing apparatus **10**. More particularly, a first end **70** of the supply tube **42** is fluidly coupled to the T-fitting **40** via a conventional nut coupler **71**, and a second end **72** of the supply tube **42** is fluidly coupled to the lower end **54** of the mounting shank **50** via a quick-connect fitting **74**. While an illustrative quick-connect fitting **74** is further detailed herein, it should be appreciated that other fluid couplings may be substituted therefor.

[0038] With further reference to FIGS. 3 and 5, the illustrative quick-connect fitting **74** includes a male portion or adapter **76** received within a female portion or receiver **78**. Conventional o-rings **79** may be supported on the adapter **76** for sealing with the receiver **78**. The male portion **76** includes an outer housing **80** defining an internal passageway **82** and receiving a check valve **84**. The housing **80** includes external threads **86** cooperating with internal threads **88** of the mounting shank **50**.

[0039] The receiver **78** of the quick-connect fitting **74** includes a receiver housing **90** receiving an inner core **92**. An outer surface **93** of the receiver housing **90** may include gripping elements (e.g., texture, knurling, annular rings, etc.) to facilitate manipulation by a user. A spring **94** is positioned between the receiver housing **90** and the inner core **92**. A valve **95** is supported by the inner core **92** and is configured to be opened when engaged by the outer housing **80** of the male portion **76**. Retention balls **96** are supported by the inner core **92** and are configured to be received within an annular slot **98** of the male portion **76**. A barbed fitting **100** on the receiver housing **90** retains the second end **72** of the supply tube **42** to the receiver **78**.

[0040] With further references to FIGS. 2-4, a drain bowl **106** is illustratively coupled to the upper end **56** of the mounting shank **50**. The drain bowl **106** includes a lower wall or base **108** and an upwardly extending wall **110**. Illustratively, the drain bowl **106** is contoured for the collection and disposal of debris and wastewater. The upper end **56** of the mounting shank **50** extends through an opening **112** of the base **108**. A coupling nut **114** includes internal threads **116** that couple with upper external threads **118** of the mounting shank **50**, and an upper flange **120** defining an opening **122**. An o-ring **124** is supported by an upper flange **126** of the mounting shank **50** and received within the opening **112** of the base **108** to seal the mounting shank **50** with the drain bowl **106**.

[0041] A drain channel **128** is in fluid communication with the drain bowl **106**. The illustrative drain channel **128** may extend through the upwardly extending wall **110**. Alternatively, the drain channel **128** may be defined independent from the upwardly extending wall **110**. As shown in FIG. 1, the drain channel **128** is configured to provide fluid communication between the drain bowl **106** and the sink basin **14** supported by the sink deck **12**. The drain channel **128** illustratively includes a channel lower wall or base **130** and opposing side walls **132** and **134**. In an installed condition of the vessel rinsing apparatus **10**, the drain bowl **106** is positioned above the mounting deck **12**, at least a portion of the drain channel **128** is positioned above the sink basin **14**, and the drain channel

**128** is configured to receive fluid from the drain bowl **106** and discharge the fluid above and into the sink basin **14**. The base **108** of the drain bowl **106** may be inclined downwardly toward the drain channel **128**. Likewise, the base **130** of the drain channel **128** may be inclined downwardly toward the sink basin **14**.

[0042] A fluid discharge control **140** is operably coupled to the drain bowl **106** and to a valve **142**. As further detailed herein, the fluid discharge control **140** includes a trigger **144** configured to be engaged by a vessel **146** (e.g., a cup, glass, bottle or other container) for activating water flow through a plurality of outlets **148** and into an interior **149** of the vessel **146** to clean an internal surface **150** (FIGS. 6A and 7A).

[0043] With reference to FIGS. 4, 6B and 7B, the illustrative valve **142** includes a fixed valve member **152**, illustratively a diverter body, received within the mounting shank **50**. An o-ring **154** is supported by the diverter body **152** and provides a seal with an inner surface **156** of the mounting shank **50**. A center opening **158** extends through the diverter body **152**. A valve seat **160** is supported by a lower surface **162** of the diverter body **152**. A valve stem **164** is slidably received within the opening **158** of the diverter body **152**.

[0044] More particularly, the valve stem **164** includes a shaft or pin **166** extending along the longitudinal axis **52** between a lower end **168** and an upper end **170**. A retaining flange **172** is positioned intermediate the lower end **168** of the valve pin **166** and the upper end **170**. A valve seal **174**, illustratively an o-ring, is received within an annular groove **176** adjacent to the retaining flange **172**. The mounting shank **50** includes an inwardly extending support wall **178** including a center opening **180** intermediate water flow openings **181** (FIGS. 6B, 7B and 12). The lower end **168** of the valve pin **166** of the valve stem **164** is slidably received within the center opening **180** of the support wall **178**. The upper end **170** of the valve pin **166** of the valve stem **164** is slidably received within the center opening **158** of the diverter body **152**.

[0045] The valve stem **164** is movable along the longitudinal axis **52** between a closed position and an open position. In the closed position, the valve seal **174** engages the valve seat **160** (FIG. 6A). In the open position, the valve seal **174** is spaced apart from the valve seat **160** (FIG. 7A). A spring **182** biases the valve stem **164** upwardly toward the closed position as shown in FIGS. 6A and 6B.

[0046] The fluid discharge control **140** is operably coupled to the valve **142**. The fluid discharge control **140** illustratively includes a central body **184** including a cylindrical side wall **186** defining a second fluid passageway **188**. A plurality of radial openings **190** are received within the side wall **186** and are in fluid communication with the second fluid passageway **188**. The second fluid passageway **188** is in selective fluid communication with the first fluid passageway **66** of the mounting shank **50** via operation of the valve **142** between the closed position (FIGS. 6A and 6B) and the open position (FIGS. 7A and 7B).

[0047] The central body **184** extends along the longitudinal axis **52** and is at least partially disposed within the mounting shank **50**. A mounting flange **192** is supported by the central body **184**. A faceplate **194** is threadably coupled to an upper end **196** of the central body **184**. An o-ring **197** is illustratively supported on the central body **184** adjacent the flange **192** and provides a seal between the central body **184** and the faceplate **194**. An engagement member **198** is coupled to a lower end **200** of the central body **184**. More particularly, the engagement member **198** illustratively includes a fastener having an externally threaded shaft **202** and a head **203**. The threaded shaft **202** illustratively engages with internal threads **204** of the central body **184**. A gasket **206**, illustratively a lip seal, is supported by the upper end **196** of the central body **184**.

[0048] A sprayface **210** is illustratively supported by the upper end **196** of the central body **184** and includes a plurality of outlets or nozzles **148** in fluid communication with the second fluid passageway **188**. Illustratively, the nozzles **148** include a center nozzle **148a** coaxially aligned with the center axis **52**, a second or intermediate set of nozzles **148b** positioned radially outwardly from the center nozzle **148a**, and a third or outer set of nozzles **148c** positioned radially outwardly from the intermediate set of nozzles **148b**.

[0049] The illustrative sprayface **210** receives and is coupled to the faceplate **194**. The nozzles **148** are illustratively oriented in different angular orientations from the longitudinal axis **52** to direct discharged water to different internal surfaces **150** of the vessel **146** (FIG. 7A).

[0050] More particularly, the illustrative sprayface **210** includes a single center nozzle **148a**, four circumferentially spaced nozzles **148b**, and four circumferentially spaced nozzles **148c**. The nozzle **148a** is configured to direct water generally upwardly along the center axis **52** (90 degrees from a plane defined by the trigger **144**), the nozzles **148b** are configured to direct water substantially 35 degrees from the center axis **52** (55 degrees from the plane defined by the trigger **144**), and the nozzles **148c** are configured to direct water substantially 70 degrees from the center axis **52** (20 degrees from the plane defined by the trigger **144**). The orientation and positioning of the nozzles **148** are configured to effectively clean a variety of vessels and prevent splashing.

[0051] With reference to FIGS. 6B, 7B, 11A-11C and 13, the illustrative fluid discharge control **140** further includes a seal setter or rotatable drive **220** operably coupled to the central body **184**. The rotatable drive **220** includes a body **222** having an upper cylindrical wall **222** supporting a plurality of radially outwardly extending pins **224**. A lower flange or lip **225** of the body **222** is secured between the lower end **200** of the central body **184** and the head **203** of the engagement member **198**. An engagement surface **226** is supported by the rotatable drive **220** and is configured to move the valve stem **164** toward the open position when the trigger **144** is in the lowered position. Illustratively, the engagement surface **226** is defined by the head **203** of the engagement member **198**.

[0052] With reference to FIGS. 4 and 9, the illustrative fluid discharge control **140** further includes a cover or retainer **230** operably coupled to the rotatable drive **220**. The cover **230** includes a body **232** received within the mounting shank **50**. An o-ring **234** is supported by the body **232** and seals with the inner surface **156** of the mounting shank **50**. The cover **230** further includes a ramp surface **236** configured to engage the pins **224** of the rotatable drive **220**. The body of the cover **230** includes a plurality of axially extending legs **237** defining a plurality of circumferentially spaced slots **238** receiving the pins **224** of the rotatable drive **220**. As further detailed herein, biasing of the trigger **144** toward the raised position causes the trigger **144** to rotate toward a rest or first rotational position.

[0053] Illustratively, the trigger **144** is operably coupled to the central body **184**. The trigger **144** illustratively includes a frame **240** having a plurality of radially outwardly extending arms **242** extending radially outwardly from a central hub **244**. A plurality of resilient inserts **246** are supported by upper channels **248** of the arms **242**. The inserts **246** are illustratively formed of a polymer, such as an elastomer or silicone.

[0054] The trigger **144** is supported for axial movement along the longitudinal axis **52** between a raised position and a lowered position. A spring **250** biases the trigger **144** toward the raised position. The trigger **144** is also supported for rotational movement about the longitudinal axis **52** between a first rotational position (FIGS. 6A-6C) and a second rotational position (FIGS. 7A-7C). According to illustrative embodiments, the second rotational position of the trigger **144** is angularly offset from the first rotational position by between 15 degrees and 45 degrees. Illustratively, the angular offset is approximately 30 degrees.

[0055] The trigger **144** is biased toward the raised position and the first rotational position (FIGS. 6A-6C). The spring **250** biases the rotatable drive **220** upwardly toward the raised position, wherein engagement of the pins **224** with the ramp surfaces **236** cause rotation of the rotatable drive **220** toward the first rotational position. Moving the trigger **144** to the lowered position is configured to move the valve seal **174** away from the valve seat **160**, thereby causing the valve **142** to be in the open position (FIGS. 7A-7C). In the closed position, the seal **174** of the valve stem **164** engages the valve seat **160** of the diverter body **152**, thereby preventing water flow from the first fluid passageway **166** of the mounting shank **50** to the second fluid passageway **188** of the central body **184**. In the open position, the seal **174** is spaced apart from the valve seat **160** such that the first

fluid passageway **166** of the mounting shank **50** is in fluid communication with the second fluid passageway **188** of the central body **184**, and thereby the outlets **148** of the sprayface **210**.

[0056] With reference to FIGS. **11A-12**, a lock **260** is configured to prevent axial movement of the trigger **144** to the lowered position when the trigger **144** is in a first rotational position. The lock **260** includes a circumferential locking surface **262** and an axial slot **264** defined by the inner surface **156** of the mounting shank **50**. A ramp surface **266** connects the locking surface **262** with the axial slot **264**. The pins **224** of the rotatable drive **220** are configured to engage the circumferential locking surface **262** when the trigger **144** is in the first rotational position, and received within the axial slot **264** when the trigger **144** is in the second rotational position.

[0057] With reference to FIGS. **6A-7C** and **11A-11C**, operation of the illustrative vessel rinsing apparatus **10** is further detailed. FIG. **6A** shows the glass **146** positioned above the trigger **144** of the vessel rinsing assembly **10**. The trigger **144** is in the raised position and the first rotational position. As such, the valve **142** is in the closed position such that no water flows through the outlets **148** of the sprayface **210**.

[0058] FIG. **7A** shows the glass **146** depressing the trigger **144** of the vessel rinsing assembly **10**. More particularly, a user initially rotates the trigger **144** is rotated clockwise by approximately **30** degrees to the second rotational position thereby axially unlocking the trigger **144** (as represented by arrow **270**). The trigger **144** is then depressed by the user pressing down on the glass **146** to the lowered position (as represented by arrow **272**). As such, the valve **142** is in the open position such that water flows through the outlets **148** of the sprayface **210**.

[0059] While FIG. **1** shows the vessel rinsing apparatus **10** supported above the upper surface **64** of the sink deck **12**, it should be appreciated that other mounting arrangements may be substituted therefor. In the illustrative embodiment of FIGS. **15** and **16**, the vessel rinsing apparatus **10** may be integrated into the sink deck **12** and recessed below the upper surface **64** thereof.

[0060] The sink **300** illustratively includes a first or main sink basin **302** and a second or auxiliary sink basin **304**. The basins **302** and **304** extend downwardly from the upper surface **64** of the sink deck **12** and include drain openings **306** and **308**, respectively. A cover **310**, illustratively a grate, extends around the vessel rinsing apparatus **10** and includes a plurality of openings **312** to allow for the passage of water therethrough into a lower receptacle or bowl **314**.

[0061] The lower receptacle **314** illustratively extends above the second sink basin **304** and includes a drain channel **316** having a ledge **318**. Water from the lower receptacle **314** passes through the drain channel **316** over the ledge **318** and into the second sink basin **304**. The water then passes to the drain opening **308**.

[0062] Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the spirit and scope of the invention as described and defined in the following claims.

## Claims

**1.** A vessel rinsing apparatus comprising: a hollow, cylindrical mounting shank extending along a longitudinal axis, the mounting shank defining a first fluid passageway; a valve including: a valve seat supported by the mounting shank, a valve stem movable along the longitudinal axis, a valve seal supported by the valve stem, wherein the valve stem is movable between a closed position and an open position, the valve seal being biased against the valve seat in the closed position; a fluid discharge control including: a central body extending along the longitudinal axis and at least partially disposed within the mounting shank, wherein the central body includes a second fluid passageway, a sprayface supported by a first end of the central body and including at least one outlet in fluid communication with the second fluid passageway, a trigger operably coupled to the central body, wherein the trigger is movable for axial movement along the longitudinal axis between a raised position and a lowered position, wherein the trigger is movable for rotational



movement about the longitudinal axis between a first rotational position and a second rotational position, wherein the trigger is biased toward the raised position, wherein moving the trigger to the lowered position is configured to move the valve seal away from the valve seat causing the valve to be in the open position, a lock configured to prevent movement of the trigger to the lowered position when the trigger is in the first rotational position; and a drain bowl operably coupled to the mounting shank, the drain bowl defined by a base and an upwardly extending wall.

**2.** The vessel rinsing apparatus of claim 1, further comprising: a drain channel fluidly coupled to the drain bowl; and wherein in an installed condition of the rinsing apparatus, the drain bowl is positioned above the mounting deck, at least a portion of the drain channel is positioned above the sink basin, and the drain channel is configured to receive fluid from the drain bowl and discharge the fluid above and into the sink basin.

**3.** The vessel rinsing apparatus of claim 1, wherein the second rotational position of the trigger is angularly offset from the first rotational position by between 15 degrees and 45 degrees.

**4.** The vessel rinsing apparatus of claim 3, wherein the second rotational position of the trigger is angularly offset from the first rotational position clockwise by approximately 30 degrees.

**5.** The vessel rinsing apparatus of claim 1, wherein: the fluid discharge control further includes: a rotatable drive operably coupled to the central body; and an engagement surface supported by the rotatable drive and configured to move the valve stem toward the open position when the trigger is in the lowered position; and the valve further includes a fixed valve member supported within the mounting shank and defining the valve seat.

**6.** The vessel rinsing apparatus of claim 5, wherein: the lock includes: a circumferential locking surface, and an axial slot defined by the mounting shank; and the rotatable drive includes: a body, and a radially extending pin configured to engage the locking surface when the trigger is in the first rotational position, and received within the axial slot when the trigger is in the second rotational position.

**7.** The vessel rinsing apparatus of claim 6, wherein the trigger is biased toward the first rotational position.

**8.** The vessel rinsing apparatus of claim 7, wherein the fluid discharge control further includes a cover operably coupled to the rotatable drive, the cover including a ramp surface configured to engage the pin of the rotatable drive wherein biasing of the trigger toward the raised position causes the trigger to move toward the first rotational position.

**9.** The vessel rinsing apparatus of claim 1, wherein the trigger of the fluid discharge control includes a plurality of arms extending radially outwardly from the central body.

**10.** The vessel rinsing apparatus of claim 1, wherein a first spring biases the trigger toward the raised position.

**11.** The vessel rinsing apparatus of claim 10, wherein a second spring biases the valve seal against the valve seat.

**12.** The vessel rinsing apparatus of claim 1, further comprising a quick connect coupler operably coupled to a lower end of the mounting shank.

**13.** A drinkware rinsing apparatus comprising: a drain bowl defined by a base and an upwardly extending wall; a valve including: a valve seat supported by the mounting shank, a valve stem movable along the longitudinal axis, a valve seal supported by the valve stem, wherein the valve stem is movable between a closed position and an open position, the valve seal being biased against the valve seat in the closed position; and a fluid discharge control including: a central body extending along the longitudinal axis and at least partially disposed within the mounting shank, wherein the central body includes an outlet fluid passageway, a sprayface supported by a first end of the central body and including at least one outlet in fluid communication with the fluid passageway, the sprayface located substantially within the drain bowl, a trigger operably coupled to the central body, wherein the trigger is movable for axial movement along the longitudinal axis between a raised position and a lowered position, wherein the trigger is biased toward the raised

position, wherein the trigger is movable for rotational movement about the longitudinal axis between a first rotational position and a second rotational position, wherein the trigger is biased toward the first rotational position, wherein moving the trigger to the lowered position is configured to move the valve seal away from the valve seat causing the valve to be in the open position, a lock configured to prevent movement of the trigger to the lowered position when the trigger is in the first rotational position.

**14.** The drinkware rinsing apparatus of claim 13, further comprising a hollow, cylindrical mounting shank extending along a longitudinal axis and operably coupled to the drain bowl, the mounting shank defining an inlet fluid passageway.

**15.** The drinkware rinsing apparatus of claim 14, further comprising: a drain channel fluidly coupled to the drain bowl; and wherein in an installed condition of the rinsing apparatus, the drain bowl is positioned above the mounting deck, at least a portion of the drain channel is positioned above the sink basin, and the drain channel is configured to receive fluid from the drain bowl and discharge the fluid above and into the sink basin.

**16.** The drinkware rinsing apparatus of claim 13, wherein the second rotational position of the trigger is angularly offset from the first rotational position by between 15 degrees and 45 degrees.

**17.** The drinkware rinsing apparatus of claim 16, wherein the second rotational position of the trigger is angularly offset from the first rotational position clockwise by approximately 30 degrees.

**18.** The drinkware rinsing apparatus of claim 13, wherein: the fluid discharge control further includes: a rotatable drive operably coupled to the central body; and an engagement surface supported by the rotatable drive and configured to move the valve stem toward the open position when the trigger is in the lowered position; and the valve further includes a fixed valve member supported within the mounting shank and defining the valve seat.

**19.** The drinkware rinsing apparatus of claim 18, wherein: the lock includes: a circumferential locking surface, and an axial slot defined by the mounting shank; and the rotatable drive includes: a body, and a radially extending pin configured to engage the locking surface when the trigger is in the first rotational position, and received within the axial slot when the trigger is in the second rotational position.

**20.** The drinkware rinsing apparatus of claim 19, wherein the fluid discharge control further includes a cover operably coupled to the rotatable drive, the cover including a ramp surface configured to engage the pin of the rotatable drive wherein biasing of the trigger toward the raised position causes the trigger to move toward the first rotational position.

**21.** The drinkware rinsing apparatus of claim 13, wherein the trigger of the fluid discharge control includes a plurality of arms extending radially outwardly from the central body.

**22.** The drinkware rinsing apparatus of claim 13, wherein a first spring biases the trigger toward the raised position.

**23.** The drinkware rinsing apparatus of claim 22, wherein a second spring biases the valve seal against the valve seat.

**24.** A vessel rinsing apparatus comprising: a hollow, cylindrical mounting shank extending along a longitudinal axis, the mounting shank defining a first fluid passageway; a valve including: a fixed valve member supported within the mounting shank and defining a valve seat, a valve stem movable along the longitudinal axis, a valve seal supported by the valve stem, wherein the valve stem is movable between a closed position and an open position, the valve seal being biased against the valve seat in the closed position; a fluid discharge control including: a central body extending along the longitudinal axis and at least partially disposed within the mounting shank, wherein the central body includes a second fluid passageway, a sprayface supported by a first end of the central body and including at least one outlet in fluid communication with the second fluid passageway, a trigger operably coupled to the central body, wherein the trigger is movable for axial movement along the longitudinal axis between a raised position and a lowered position, wherein the trigger is movable for rotational movement about the longitudinal axis between a first rotational position and

a second rotational position, wherein the trigger is biased toward the raised position, wherein moving the trigger to the lowered position is configured to move the valve seal away from the valve seat causing the valve to be in the open position, a lock configured to prevent movement of the trigger to the lowered position when the trigger is in the first rotational position; a rotatable drive operably coupled to the central body, an engagement surface supported by the rotatable drive and configured to move the valve stem toward the open position when the trigger is in the lowered position; a drain bowl operably coupled to the mounting shank, the drain bowl defined by a base and an upwardly extending wall; a drain channel fluidly coupled to the drain bowl; and wherein in an installed condition of the rinsing apparatus, the drain bowl is positioned above the mounting deck, at least a portion of the drain channel is positioned above the sink basin, and the drain channel is configured to receive fluid from the drain bowl and discharge the fluid above and into the sink basin.

**25.** The vessel rinsing apparatus of claim 24, wherein the second rotational position of the trigger is angularly offset from the first rotational position by between 15 degrees and 45 degrees.

**26.** The vessel rinsing apparatus of claim 25, wherein the second rotational position of the trigger is angularly offset from the first rotational position clockwise by approximately 30 degrees.

**27.** The vessel rinsing apparatus of claim 24, wherein: the lock includes: a circumferential locking surface, and an axial slot defined by the mounting shank; and the rotatable drive includes: a body, and a radially extending pin configured to engage the locking surface when the trigger is in the first rotational position, and received within the axial slot when the trigger is in the second rotational position.

**28.** The vessel rinsing apparatus of claim 27, wherein the trigger is biased toward the first rotational position.

**29.** The vessel rinsing apparatus of claim 28, wherein the fluid discharge control further includes a cover operably coupled to the rotatable drive, the cover including a ramp surface configured to engage the pin of the rotatable drive wherein biasing of the trigger toward the raised position causes the trigger to move toward the first rotational position.

**30.** The vessel rinsing apparatus of claim 24, wherein the trigger of the fluid discharge control includes a plurality of arms extending radially outwardly from the central body.

**31.** The vessel rinsing apparatus of claim 24, wherein a first spring biases the trigger toward the raised position.

**32.** The vessel rinsing apparatus of claim 31, wherein a second spring biases the valve seal against the valve seat.

**33.** The vessel rinsing apparatus of claim 24, further comprising a quick connect coupler operably coupled to a lower end of the mounting shank.

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